Chapter 1

Transmission Models for Mechanistic Analysis of RSV Seasonality

1.1 Seasonal epidemics as a dynamical inference problem

Seasonal epidemics reflect the interaction between time-varying transmission and the evolving distribution of susceptibility in a population. The timing of an epidemic is therefore not determined by the transmission rate alone. Susceptible depletion, replenishment through demographic processes and waning immunity, and the state of the system at the beginning of a season jointly determine the resulting incidence trajectory. A change in epidemic timing can consequently arise without an equivalent displacement of the underlying seasonal transmission curve.

This distinction is central to the interpretation of dynamical transitions. Earn et al. [1] demonstrated that a simple nonlinear epidemic model can generate qualitatively different epidemic regimes as demographic and vaccination conditions change. Their analysis illustrates a general principle: substantial changes in observed epidemic patterns need not require equally substantial changes in the external forcing. Conversely, a change in an observed epidemic pattern does not uniquely identify the parameter or process responsible for that change.

For respiratory syncytial virus (RSV), the representation of susceptibility requires particular attention because infection does not confer lifelong sterilizing immunity. White et al. [2] compared nested models incorporating alternative assumptions about reinfection and immunity across multiple populations. Their results showed that the treatment of partial and waning immunity materially affects the ability to reproduce RSV dynamics. The susceptible–infectious–recovered–susceptible (SIRS) framework offers a parsimonious representation of recurrent infection, although its compartments should be interpreted as population-level approximations rather than a complete account of individual immune

histories.

These considerations motivate a distinction between two modelling tasks. The first is to reconstruct an observed epidemic trajectory. The second is to determine how that trajectory changes under a specified perturbation. The latter requires a calibrated dynamical model whose baseline trajectory adequately represents the features under investigation. A simulator that substantially misplaces an epidemic peak may describe the response of a mathematically valid system while providing a poor approximation to the particular transition observed in the data.

The objective of the present study is to construct such a simulator for Japanese RSV dynamics and use it to examine the mechanisms of epidemic timing shifts. Accurate reconstruction is treated as an empirical requirement for this analysis, rather than as evidence that the inferred mechanism is uniquely identified.

1.2 Existing representations of RSV transmission

1.2.1 Periodic forcing and climate-dependent transmission

A common representation of seasonal transmission uses a small number of harmonic components. For example, a positive annual transmission curve can be written as

$$\log \beta(t) = \alpha_0 + a_1 \cos\left(\frac{2\pi t}{P}\right) + b_1 \sin\left(\frac{2\pi t}{P}\right), \tag{1.1}$$

where $\beta(t) > 0$ is the transmission rate and P is the seasonal period. Adding sine and cosine terms at frequency $2/P$ introduces a semiannual component. Such representations are economical and provide interpretable frequency components, but restrict the admissible seasonal shapes.

RSV modelling has established that recurrent transmission forcing can interact with population dynamics to produce more complex incidence patterns. Pitzer et al. [3] reproduced geographical variation in RSV epidemic dynamics in the United States, including biennial patterns arising under an annually repeating transmission regime. Their analysis also connected estimated seasonal transmission characteristics with environmental variation. Thus, the periodicity of observed incidence need not coincide with the periodicity imposed on transmission.

Climate-dependent models provide a different route to specifying $\beta(t)$. Baker et al. [4] linked RSV transmission to specific humidity and precipitation across the United States and Mexico. This approach connects epidemic dynamics to measurable environmental drivers and supports projections under changing climatic conditions. Its interpretation nevertheless depends on the specified relationship between climate and transmission and on the extent to which other time-varying influences are represented.

For the present purpose, neither a low-order harmonic expansion nor a climate regression should be accepted solely because it produces a plausible seasonal curve. The relevant question is whether its restrictions preserve the epidemic timing responses that the study seeks to explain. A representation that is adequate for average seasonality may still be inadequate for a transition involving changes in peak width, asymmetry, or the relative importance of competing epidemic maxima.

1.2.2 The Japanese RSV seasonality-shift model

Park et al. [5] investigated the abrupt change in RSV seasonality in Japan using a discrete-time SIRS model with a flexible seasonal transmission function. Their transmission specification takes the multiplicative form

$$\beta_{\text{Park}}(t) = \beta_{\text{seas}}\{w(t)\}\, q(t), \tag{1.2}$$

where $w(t) \in \{1, \ldots, 52\}$ denotes seasonal week, β_{seas} is a repeating seasonal transmission profile, and $q(t)$ represents the time-varying modification associated with the pandemic period. For the Honshu specification, $q(t) = 1$ before 2020.

Importantly, the seasonal profile in this model is not restricted to a sinusoid. Separate positive transmission values are estimated for the 52 seasonal weeks, with cyclic random-walk priors linking adjacent values. Post-2019 modifications are also represented by weekly parameters subject to smoothing priors. The inferred seasonal transmission profiles are subsequently related to climatic variables. The reported semiannual transmission pattern is therefore an empirical result of their analysis rather than simply an imposed two-harmonic specification.

Park et al. showed that changes in susceptibility or transmission level can produce substantial changes in epidemic timing under the inferred seasonal regime. This establishes a mechanistic possibility and provides the immediate starting point for the present study. The outstanding issue is whether the detailed interpretation of the timing transition remains stable under alternative, empirically competitive representations of transmission.

The distinction from the present approach is consequently not "flexible" versus "inflexible" transmission. It concerns how flexibility is allocated across seasonal shape and changes between seasons, how the resulting model is estimated, and how its calibrated trajectories support subsequent perturbation analysis.

1.3 Why additional modelling flexibility is required

1.3.1 A common seasonal profile can conceal changes between seasons

A repeating seasonal transmission curve assumes that the same within-year profile applies across the historical period. Even when this curve is estimated nonparametrically, its repetition remains a substantive restriction. Under this specification, changes in historical epidemic timing must primarily be explained through the evolving compartment states and their interaction with a shared forcing function.

This restriction is scientifically useful when testing whether susceptibility dynamics alone can explain an observed transition. It is less suitable as an unquestioned assumption when the objective is to compare changes in susceptibility with changes in transmission. If transmission itself varies between seasons, forcing all historical seasons to share the same profile may alter the inferred compartment states and the subsequent interpretation of timing sensitivity.

An alternative simulator should therefore allow a controlled departure from a common seasonal profile. The departure should be sufficiently structured to distinguish changes in transmission level from changes in seasonal shape, while avoiding an unrestricted parameter for every historical observation.

1.3.2 Aggregate fit does not establish timing fidelity

The likelihood and aggregate count errors summarize discrepancies over many observations. They do not directly measure whether a model reproduces the particular timing transition of interest. A model can achieve an acceptable overall error while placing a key epidemic peak several weeks too early or too late.

Moreover, an estimated change between two peak dates can be accurate even when both dates are inaccurate. Let T_1 and T_2 denote observed peak times in two seasons, and let $\widehat{T}_1$ and $\widehat{T}_2$ denote the corresponding simulated peak times. Then

$$(\widehat{T}_2 - \widehat{T}_1) - (T_2 - T_1) = (\widehat{T}_2 - T_2) - (\widehat{T}_1 - T_1). \tag{1.3}$$

Similar endpoint errors can cancel in the estimated displacement. Consequently, a small shift error does not imply that the simulator accurately reconstructs either epidemic.

For mechanistic analysis, reconstruction must therefore be assessed at complementary levels: the weekly epidemic trajectory, the timing of individual epidemics, and the displacement between epidemics. This requirement follows from the scientific target and cannot be replaced by selecting the model with the largest likelihood alone.

1.3.3 Flexibility must remain constrained

Increasing transmission flexibility can improve reconstruction without resolving the underlying biological explanation. A time-varying transmission function may absorb discrepancies caused by omitted population structure, changes in reporting, or an inadequate representation of immunity.

The modelling objective is therefore not unrestricted interpolation. It is to identify a regularized transmission representation that reproduces the relevant epidemic features while retaining a transparent dynamical structure. The resulting transmission corrections are interpreted as effective model components. They are not assigned to specific behavioural or biological causes without additional evidence.

1.4 Spline representations as a methodological foundation

Penalized splines provide a useful compromise between a restrictive harmonic representation and a separate parameter at every time point. Eilers and Marx [6] developed a framework in which a spline basis supplies flexibility and a difference penalty controls roughness. This separates the representation of a function from the degree of smoothness imposed during estimation.

Spline methods also have an established role in dynamical inference. Ramsay et al. [7] developed generalized profiling for differential-equation models using a smoothing-based representation of trajectories. Their framework demonstrates how statistical flexibility can be combined with dynamical constraints. The present approach differs in its implementation: splines parameterize transmission, while the epidemic trajectory is generated by the specified SIRS recurrence.

A more direct precedent is the work of Bouman et al. [8], who compared Brownian-motion, B-spline, and approximate Gaussian-process representations of time-varying transmission within a Bayesian compartmental workflow. Their experiments found computational advantages for B-splines while retaining accurate reconstruction of the simulated transmission functions. Their HETTMO implementation uses an SEIR structure and does not constitute an RSV-specific SIRS model; its relevance here is methodological.

Spline representations are also present in Park et al.'s analysis: cyclic cubic splines were used to smooth inferred seasonal transmission profiles for selected counterfactual comparisons [5]. The methodological contribution sought here is therefore not the introduction of splines into epidemic modelling. It is their integration into a directly calibrated, regularized SIRS simulator with an explicit decomposition of shared seasonality and temporal deviations.

Let $B_1, \ldots, B_K$ be cyclic cubic spline basis functions and define

$$g(w) = \sum_{k=1}^{K} c_k B_k(w). \tag{1.4}$$

A normalized seasonal transmission profile is

$$\beta_{\text{seas}}(w) = \beta_0 \frac{\exp\{g(w)\}}{P^{-1} \sum_{v=1}^{P} \exp\{g(v)\}}, \qquad P = 52, \tag{1.5}$$

so that the arithmetic mean over the seasonal grid equals β_0. The exponential link ensures positivity, and cyclic boundary conditions join the end and beginning of the seasonal cycle smoothly.

The proposed transmission decomposition is

$$\beta(t) = \beta_{\text{seas}}\{w(t)\}\, m(t), \tag{1.6}$$

where $m(t) > 0$ represents structured temporal departures. In the historical component of M2,

$$\log m(t) = a_{s(t)} + b_{s(t),1} \widetilde{\sin}\{w(t)\} + b_{s(t),2} \widetilde{\cos}\{w(t)\}, \tag{1.7}$$

where $s(t)$ indexes epidemic seasons and the two harmonic functions are centered over the seasonal grid. The coefficient a_s controls the log-scale level correction, while $b_{s,1}$ and $b_{s,2}$ describe a low-dimensional modification of seasonal shape. A separate smooth temporal correction is used for the post-2019 period.

This decomposition makes the intended interpretation explicit: a shared seasonal profile is distinguished from season-specific departures. Centering provides a parameterization convention; it does not, by itself, establish biological identifiability. In particular, zero-mean shape corrections on the logarithmic scale do not imply an unchanged arithmetic mean transmission after exponentiation.

1.5 Bayesian inference and penalized maximum likelihood

The transmission representation and the inferential framework are separate modelling choices. Splines can be estimated in either a Bayesian or a likelihood-based framework, as illustrated by the Bayesian spline model of Bouman et al. [8]. Likewise, Bayesian estimation does not inherently imply a particular degree of model flexibility.

Let ϑ collect the transmission coefficients, initial conditions, biological parameters being estimated, and observation parameters. Given a simulated weekly infection incidence

$C_t(\vartheta)$, the observation model is

$$Y_t \mid \vartheta \sim \text{NegBin}\Big(\mu_t(\vartheta), \phi\Big), \qquad \mu_t(\vartheta) = \rho C_t(\vartheta), \tag{1.8}$$

with

$$\text{Var}(Y_t \mid \vartheta) = \mu_t(\vartheta) + \frac{\mu_t(\vartheta)^2}{\phi}. \tag{1.9}$$

Here Y_t denotes reported cases, ρ is the reporting multiplier, and ϕ controls overdispersion. This observation layer separates reported cases from simulated infections.

Writing the log-likelihood over included observations as $\ell(\vartheta)$, unpenalized maximum likelihood estimates

$$\widehat{\vartheta}_{\text{ML}} = \arg\max_{\vartheta \in \Theta} \ell(\vartheta), \tag{1.10}$$

where Θ is the admissible parameter space. The proposed spline model instead uses penalized maximum likelihood:

$$\widehat{\vartheta}_{\text{PML}} = \arg\max_{\vartheta \in \Theta} \left\{ \ell(\vartheta) - \sum_{j=1}^{J} \lambda_j \mathcal{P}_j(\vartheta) \right\}. \tag{1.11}$$

The penalties $\mathcal{P}_j$ constrain roughness and the magnitude or variation of temporal corrections, with strengths $\lambda_j \geq 0$.

In contrast, Bayesian inference combines the likelihood with a prior density $\pi(\vartheta)$:

$$p(\vartheta \mid Y) \propto \exp\{\ell(\vartheta)\}\pi(\vartheta). \tag{1.12}$$

Park et al. use posterior sampling to represent uncertainty in the fitted model [5]. A posterior distribution supports uncertainty propagation through subsequent simulations, whereas a penalized likelihood fit initially supplies a single calibrated parameter vector.

The distinction is not that penalized likelihood is free of regularizing assumptions. When a penalty corresponds to a negative log-prior, the penalized optimum can coincide with a maximum a posteriori estimate under a matching parameterization. However, computing this optimum is not equivalent to exploring the posterior distribution. Uncertainty assessment for the penalized fit requires additional procedures, such as profiling or bootstrap refitting.

The choice of penalized likelihood in the present study is motivated by the intended computational workflow: repeated calibration, controlled perturbations, sensitivity calculations, and comparisons of transmission representations. Its value must be assessed through empirical fit and computational performance, rather than through a general claim of superiority over Bayesian inference.

1.6 Requirements for a simulator of timing shifts

1.6.1 Computational efficiency for repeated analysis

For a fixed-dimensional SIRS state and an available sequence of transmission values, a forward simulation over T observations requires a number of state updates proportional to T. This property is shared by both transmission representations. A spline parameterization therefore does not, by itself, change the asymptotic order of the forward recurrence.

Its computational advantage may instead arise from a smaller parameter representation, structured smoothness, and the number and cost of model evaluations required during estimation. Total runtime also depends on optimization starts, derivative calculations, sampler settings, numerical tolerances, and parallelization. Accordingly, comparisons must distinguish fitting time from simulation time and identify the inferential output being produced.

A penalized point estimate and a posterior sample are different computational products. A shorter fitting time for the former can make repeated mechanistic analyses more practical, but should not be interpreted as producing equivalent uncertainty information at lower cost.

1.6.2 Interpretability through explicit perturbations

The proposed decomposition supplies coordinates for controlled experiments. Transmission level can be perturbed while retaining seasonal shape; seasonal shape can be modified while retaining the underlying compartment model; and initial susceptibility can be varied under a fixed transmission trajectory.

These operations allow competing explanations to be compared within a common dynamical system. Their interpretability comes from the specification of what changes and what is held fixed, rather than from the use of a particular estimation algorithm. The biological interpretation of any fitted correction remains conditional on the observation and compartmental assumptions.

1.6.3 Mathematical formulation of peak movement

Let $z(t;\eta)$ denote a smooth epidemic trajectory under a scalar perturbation η, and let $t^\star(\eta)$ denote a local maximum. If the maximum is interior and nondegenerate, then

$$\partial_t z\{t^\star(\eta);\eta\} = 0, \qquad \partial_{tt} z\{t^\star(\eta);\eta\} < 0. \tag{1.13}$$

Implicit differentiation gives

$$\frac{dt^\star}{d\eta} = -\frac{\partial_{t\eta} z\{t^\star(\eta);\eta\}}{\partial_{tt} z\{t^\star(\eta);\eta\}}. \tag{1.14}$$

This expression distinguishes the local movement of a peak from a change in which of several peaks is globally dominant. It also shows why relatively flat peaks can exhibit large timing sensitivity.

Equation (1.14) is a local statement for a smooth trajectory. A peak defined by the maximum of weekly observations is a discrete statistic and can jump between weeks or between distinct local maxima. Consequently, smooth branch movement, exchanges in peak dominance, and discrete observation effects must be distinguished in numerical experiments. Smooth seasonal forcing alone does not make the global peak-time functional differentiable.

These requirements define the role of the proposed spline-SIRS model. It is intended to provide an empirically calibrated and computationally tractable system in which alternative explanations of timing shifts can be formulated and tested. Its adequacy is to be established through reconstruction of epidemic trajectories and timing, numerical diagnostics, comparisons with alternative transmission representations, and temporal prediction under explicit extrapolation rules.

Chapter 2

SF-SIRS and the Mathematical Structure of Epidemic Peak Shifts

2.1 Model specification and analytical scope

We introduce the Spline–Fourier SIRS model (SF-SIRS), a finite-dimensional transmission model embedded in a weekly susceptible–infectious–recovered–susceptible system. The model combines a cyclic spline representation of baseline seasonality with season-specific Fourier corrections and a time-varying spline correction. This construction permits transmission patterns to vary across seasons while retaining an explicit mathematical representation.

All analyses in this chapter concern Honshu during 2013–2023. Season-specific analyses use windows extending from epidemiological week 27 of year y to week 26 of year $y+1$, for $y = 2013, \ldots, 2022$. The first and last partial seasons are treated separately. The seasonal forcing coordinate has period 52, whereas observation windows follow the recorded weekly panel; these two time conventions are distinguished when a calendar year contains week 53.

The mathematical results below characterize the fitted dynamical system and its perturbations. The relative frequency of different shift mechanisms, the ordering of transmission and susceptibility sensitivities, and the magnitude of initial-state effects are empirical hypotheses. They are not assumed consequences of the model specification.

2.1.1 A finite-dimensional transmission function

Let $w_t \in \{1, \ldots, 52\}$ denote the seasonal forcing index at weekly transition t, and put $x_t = w_t - 1$. For a fixed cyclic cubic B-spline basis $B_1, \ldots, B_K$, define

$$g(x; \mathbf{c}) = \sum_{j=1}^{K} c_j B_j(x), \qquad \sum_{j=1}^{K} c_j = 0. \tag{2.1}$$

The normalized baseline transmission function is

$$\beta_{\text{base}}(x) = \beta_0 \frac{\exp\{g(x;\mathbf{c})\}}{52^{-1}\sum_{\ell=0}^{51}\exp\{g(\ell;\mathbf{c})\}}, \qquad \beta_0 > 0. \tag{2.2}$$

Consequently,

$$\frac{1}{52}\sum_{\ell=0}^{51}\beta_{\text{base}}(\ell) = \beta_0. \tag{2.3}$$

The normalization separates the baseline transmission level from its seasonal shape.

Let $s(t)$ identify the epidemic season containing observation t. For the historical correction block, corresponding to 2013–2019, write

$$h_t = a_{s(t)} + b_{s(t)}\psi_{\sin}(w_t) + d_{s(t)}\psi_{\cos}(w_t), \tag{2.4}$$

where

$$\psi_{\sin}(w) = \sin(2\pi w/52) - \frac{1}{52}\sum_{\ell=1}^{52}\sin(2\pi\ell/52), \tag{2.5}$$

$$\psi_{\cos}(w) = \cos(2\pi w/52) - \frac{1}{52}\sum_{\ell=1}^{52}\cos(2\pi\ell/52). \tag{2.6}$$

The level coefficients satisfy $\sum_{s\in\mathcal{S}_H} a_s = 0$, where $\mathcal{S}_H$ is the set of historical seasons assigned a correction. Corrections are zero for unassigned seasons. The coefficients b_s and d_s modify within-season shape without requiring the entire transmission function to belong to a low-order Fourier family.

For the later correction block, corresponding to 2020–2023, let $C_1, \ldots, C_L$ be the fixed temporal spline basis and define

$$q_t = \sum_{\ell=1}^{L}\xi_\ell C_\ell(t). \tag{2.7}$$

Writing H_t and P_t for the indicators of the historical and later blocks, respectively, the effective transmission rate is

$$\boxed{\beta_t(\boldsymbol{\theta}) = \beta_{\text{base}}(x_t)\exp\{H_t h_t + P_t q_t\}.} \tag{2.8}$$

Here $\boldsymbol{\theta} = (\beta_0, \mathbf{c}, \mathbf{a}, \mathbf{b}, \mathbf{d}, \boldsymbol{\xi})$, subject to the stated constraints, is a finite-dimensional parameter vector.

Equation (2.8) is the defining transmission representation of SF-SIRS. Once the bases and calendar assignments are fixed, transmission and its parameter derivatives are explicit functions of finitely many coordinates. This structure supports reproducible perturbations, tangent recursions, peak-branch continuation, and comparisons between distinct

sources of timing variation.

The raw season-specific correction can change at season boundaries. Thus, smoothness with respect to model parameters does not imply global smoothness with respect to calendar time. A boundary-blended correction can be examined as a robustness specification, with its smoothing rule held fixed during perturbation analysis.

2.1.2 Weekly SIRS dynamics

Let S_t, I_t, R_t denote compartment counts after transition t, and let N be the fixed population denominator. The parameters $\mu > 0$, $\gamma \geq 0$, and $\delta \geq 0$ represent demographic turnover, recovery, and loss of immunity, respectively. An additive infectious-pressure term $\omega \geq 0$ enters the force of infection:

$$\lambda_t = \beta_t \frac{I_{t-1} + \omega}{N}. \tag{2.9}$$

Define the weekly compartment outflows

$$U_t^S = S_{t-1}\left[1 - \exp\{-(\lambda_t + \mu)\}\right], \tag{2.10}$$

$$U_t^I = I_{t-1}\left[1 - \exp\{-(\gamma + \mu)\}\right], \tag{2.11}$$

$$U_t^R = R_{t-1}\left[1 - \exp\{-(\delta + \mu)\}\right]. \tag{2.12}$$

The infection, recovery, and immunity-loss flows are

$$J_t = \frac{\lambda_t}{\lambda_t + \mu} U_t^S, \tag{2.13}$$

$$G_t = \frac{\gamma}{\gamma + \mu} U_t^I, \tag{2.14}$$

$$L_t = \frac{\delta}{\delta + \mu} U_t^R. \tag{2.15}$$

The state update is

$$\boxed{\begin{aligned} S_t &= S_{t-1} - U_t^S + L_t + \mu N, \\ I_t &= I_{t-1} + J_t - U_t^I, \\ R_t &= R_{t-1} + G_t - U_t^R. \end{aligned}} \tag{2.16}$$

We write this recurrence compactly as

$$X_t = F_t(X_{t-1}; \boldsymbol{\vartheta}), \qquad X_t = (S_t, I_t, R_t)^\top, \tag{2.17}$$

where $\boldsymbol{\vartheta}$ collects transmission and epidemiological parameters.

The recurrence is used directly in all theoretical perturbations. In particular, it is not replaced by a continuous-time SIRS approximation. The saved total compartment mass

at a restart time need not equal N exactly under this discrete demographic update.

2.1.3 Observation model and estimation

Let Y_t be the reported weekly case count and let $\rho \in (0, 1]$ be the reporting proportion. The observation model is

$$Y_t \mid X_{t-1} \sim \mathrm{NB}_2(m_t, \phi), \qquad m_t = \rho J_t, \qquad \mathrm{Var}(Y_t \mid X_{t-1}) = m_t + \frac{m_t^2}{\phi}, \tag{2.18}$$

with $\phi > 0$. Parameters are estimated by constrained penalized likelihood:

$$\widehat{\boldsymbol{\vartheta}} \in \arg\min_{\boldsymbol{\vartheta} \in \Theta} \left\{ -\sum_{t \in \mathcal{T}_{\mathrm{lik}}} \log p(Y_t \mid m_t, \phi) + \mathcal{P}(\boldsymbol{\vartheta}) \right\}. \tag{2.19}$$

The penalty controls cyclic spline roughness, temporal correction roughness, and the size and between-season variation of historical corrections. The initialization row is excluded from the likelihood.

Mechanistic perturbations are evaluated conditionally on the selected fitted parameter vector. Because ρ is constant and positive, m_t and J_t have the same deterministic peak locations. Observed counts Y_t, however, need not attain their maximum at the deterministic peak.

2.2 Peak timing and its mathematical representation

For a season window $\mathcal{W}_y$, define the selected weekly peak

$$k_y(\mathbf{p}) = \min \arg\max_{t \in \mathcal{W}_y} J_t(\mathbf{p}), \tag{2.20}$$

where $\mathbf{p}$ denotes a perturbation vector. The earliest maximizer is selected when ties occur.

The integer-valued map k_y is generally locally constant and can jump when two sampled values exchange order. Its finite differences therefore do not, by themselves, estimate a smooth peak-time derivative.

For local timing analysis, let k be an interior sampled peak with available neighbors and define

$$A(\mathbf{p}) = J_{k-1}(\mathbf{p}) - J_{k+1}(\mathbf{p}), \tag{2.21}$$

$$B(\mathbf{p}) = J_{k-1}(\mathbf{p}) - 2J_k(\mathbf{p}) + J_{k+1}(\mathbf{p}). \tag{2.22}$$

When $B < 0$ and the fitted vertex belongs to the accepted local neighborhood, the

quadratic peak-time reconstruction is

$$\tau(\mathbf{p}) = k + \frac{A(\mathbf{p})}{2B(\mathbf{p})}. \tag{2.23}$$

This construction provides a differentiable timing functional on a fixed eligible stencil. It does not assert that the weekly epidemic trajectory is itself a continuous-time solution.

For a general smooth reconstruction $f(t, \mathbf{p})$, a nondegenerate local peak branch satisfies

$$f_t(\tau(\mathbf{p}), \mathbf{p}) = 0, \qquad f_{tt}(\tau(\mathbf{p}), \mathbf{p}) < 0. \tag{2.24}$$

All derivative statements are restricted to regions where the branch, observation window, and reconstruction remain eligible.

2.3 Hypothesis 1: mechanisms of peak change

2.3.1 Branch displacement and dominance exchange

Let p be a scalar perturbation coordinate. Suppose that a smooth reconstruction possesses local peak branches $\tau_1(p), \ldots, \tau_M(p)$, with heights

$$H_j(p) = f(\tau_j(p), p). \tag{2.25}$$

The selected peak is determined jointly by the branch locations and their relative heights.

Proposition 1: local displacement. If f is twice continuously differentiable near (τ_0, p_0), with $f_t(\tau_0, p_0) = 0$ and $f_{tt}(\tau_0, p_0) < 0$, then there is a locally unique differentiable peak branch satisfying

$$\boxed{\frac{d\tau}{dp} = -\frac{f_{tp}(\tau(p), p)}{f_{tt}(\tau(p), p)}.} \tag{2.26}$$

If the branch also retains a strict global height advantage, the selected peak follows this branch locally. Its location can therefore change without a dominance exchange.

Proposition 2: dominance exchange. Suppose two distinct nondegenerate branches satisfy

$$H_1(p_0) = H_2(p_0), \qquad \tau_1(p_0) \neq \tau_2(p_0), \tag{2.27}$$

and both exceed all remaining candidate maxima by a positive margin. If

$$\frac{d}{dp}\{H_1(p) - H_2(p)\}\Big|_{p=p_0} \neq 0, \tag{2.28}$$

then the globally selected branch changes across p_0. The resulting peak-time jump is determined by the separation between the competing branches.

Branch birth or loss can occur when nondegeneracy fails. A selected maximum can also reach a window boundary, where an interior-peak derivative is no longer applicable. These cases are retained as separate outcomes rather than forced into either displacement or dominance exchange.

2.3.2 Observation-induced differences

Let $\mathcal{O}_\eta$ denote an observation operator indexed by η, such as temporal aggregation, reporting delay, or time-varying reporting intensity. The associated observed-mean peak is

$$k_y^{\mathrm{obs}}(\mathbf{p}, \eta) = \min \arg\max_{t \in \mathcal{W}_y} [\mathcal{O}_\eta J(\mathbf{p})]_t. \tag{2.29}$$

A change in k_y^{obs} at fixed J is an observation-induced timing change. Under the fitted constant-reporting specification, $\mathcal{O}_\rho J = \rho J$, such a deterministic change cannot be generated by varying $\rho > 0$ alone.

A sampled peak change, a truncated observation window, and a change in the observation operator are therefore distinct objects.

Hypothesis H1. Within the prespecified SF-SIRS perturbation design for Honshu, peak displacement occurs without dominance exchange and accounts for a substantial share of detected timing-change events.

To formulate a majority claim, let $\mathcal{E}_y$ be the parent-grid timing events in season y, and let $\mathcal{D}_y \subseteq \mathcal{E}_y$ contain events resolved as pure displacement. Define

$$\pi_y^D = \frac{|\mathcal{D}_y|}{|\mathcal{E}_y|}, \qquad |\mathcal{E}_y| > 0. \tag{2.30}$$

A season-averaged majority hypothesis is

$$\frac{1}{|\mathcal{Y}_E|} \sum_{y \in \mathcal{Y}_E} \pi_y^D > \frac{1}{2}, \qquad \mathcal{Y}_E = \{y : |\mathcal{E}_y| > 0\}. \tag{2.31}$$

Mixed and unresolved events remain in the denominator. Their alternative allocations yield sensitivity bounds.

Event frequency is distinguished from cumulative absolute peak movement. A small number of dominance exchanges may produce large timing changes even when displacement is more frequent. Neither quantity is interpreted as a population probability independent of the chosen perturbation domain and grid.

2.4 Hypothesis 2: relative sensitivity to transmission and susceptibility

Let r_y be the saved state immediately preceding the first affected transition of season y, and define

$$M_{r_y} = S_{r_y} + I_{r_y} + R_{r_y}. \tag{2.32}$$

Consider matched log-scale perturbations

$$S_{r_y}(u) = S_{r_y} e^u, \tag{2.33}$$
$$I_{r_y}(u) = I_{r_y}, \tag{2.34}$$
$$R_{r_y}(u) = M_{r_y} - S_{r_y} e^u - I_{r_y}, \tag{2.35}$$
$$\beta_t(v) = \beta_t e^v, \qquad t \in \mathcal{W}_y. \tag{2.36}$$

Only feasible initial states are admitted. The susceptibility perturbation changes the initial state; the transmission perturbation changes the entire effective transmission schedule within the target window.

Let $\tau_y(u, v)$ denote the eligible reconstructed peak time. Define

$$E_{S,y} = \left.\frac{\partial \tau_y}{\partial u}\right|_{(0,0)}, \qquad E_{\beta,y} = \left.\frac{\partial \tau_y}{\partial v}\right|_{(0,0)}. \tag{2.37}$$

These sensitivities have units of weeks per unit log change. For a small proportional increase ϵ, the timing response is approximately $E_{q,y}\epsilon$.

Proposition 3: sensitivity comparison on a common branch. Under the conditions of Proposition 1,

$$E_{S,y} = -\frac{f_{tu}}{f_{tt}}, \qquad E_{\beta,y} = -\frac{f_{tv}}{f_{tt}}, \tag{2.38}$$

with all derivatives evaluated at the baseline peak. Consequently,

$$|E_{\beta,y}| > |E_{S,y}| \quad \Longleftrightarrow \quad |f_{tv}| > |f_{tu}|. \tag{2.39}$$

The two responses share the same baseline peak curvature, but differ in how their perturbations alter the local temporal slope.

For the quadratic reconstruction,

$$\boxed{E_{q,y} = \frac{BA_q - AB_q}{2B^2}, \qquad q \in \{u, v\},} \tag{2.40}$$

on a fixed eligible stencil.

Hypothesis H2. Transmission produces the larger absolute local timing response in a majority of eligible Honshu seasons:

$$\frac{1}{|\mathcal{Y}_{\text{elig}}|}\sum_{y\in\mathcal{Y}_{\text{elig}}}\mathbf{1}\{|E_{\beta,y}|>|E_{S,y}|\}>\frac{1}{2}. \tag{2.41}$$

This hypothesis does not impose a universal inequality. The signs of $E_{S,y}$ and $E_{\beta,y}$ are reported separately: negative values indicate advancement, and positive values indicate delay.

Numerical estimates use matched central differences,

$$\widehat{E}_{q,y}(h)=\frac{\tau_y(h\mathbf{e}_q)-\tau_y(-h\mathbf{e}_q)}{2h}, \tag{2.42}$$

with branch, stencil, feasibility, and step-size diagnostics. Different step sizes are numerical checks rather than independent statistical replicates.

2.5 Hypothesis 3: initial infectious-state memory

The SIRS state contains information beyond susceptibility. Even at fixed initial S and fixed transmission schedule, the initial infectious population can alter subsequent incidence, susceptible depletion, and peak timing.

Define the feasible perturbation

$$S_{r_y}(z)=S_{r_y}, \tag{2.43}$$
$$I_{r_y}(z)=I_{r_y}e^z, \tag{2.44}$$
$$R_{r_y}(z)=M_{r_y}-S_{r_y}-I_{r_y}e^z. \tag{2.45}$$

The transmission schedule is unchanged. Subsequent susceptibility evolves according to Equation (2.16); only its initial value is fixed.

Let

$$E_{I,y}=\left.\frac{\partial\tau_y(z)}{\partial z}\right|_{z=0}. \tag{2.46}$$

Proposition 4: propagation of initial-state sensitivity. Let

$$Z_t=\left.\frac{\partial X_t(z)}{\partial z}\right|_{z=0}. \tag{2.47}$$

Then

$$Z_{r_y}=(0,I_{r_y},-I_{r_y})^\top,\qquad Z_t=D_X F_t\,Z_{t-1}. \tag{2.48}$$

More generally, for a perturbation q entering both the state and the transition map,

$$Z_t^{(q)} = D_X F_t \, Z_{t-1}^{(q)} + \partial_q F_t. \tag{2.49}$$

Thus, an initial infectious-state perturbation propagates through the full trajectory even when transmission is held fixed.

For an eligible reconstructed peak,

$$E_{I,y} = -\frac{f_{tz}}{f_{tt}}, \tag{2.50}$$

or, for the quadratic reconstruction,

$$E_{I,y} = \frac{BA_z - AB_z}{2B^2}. \tag{2.51}$$

Hypothesis H3. Initial infectious-state variation provides an additional peak-timing mechanism:

$$E_{I,y} \neq 0 \tag{2.52}$$

for at least one eligible season, with its prevalence and magnitude evaluated across the Honshu seasonal panel. The stronger directional hypothesis $E_{I,y} < 0$, corresponding to advancement under increased initial infectious prevalence, is assessed separately.

Finite changes are summarized by

$$\Delta\tau_{I,y}(a) = \tau_y(\log a) - \tau_y(-\log a), \qquad a > 1. \tag{2.53}$$

The corresponding initial infectious counts are I_{r_y}/a and aI_{r_y}, whose ratio is a^2. A finite contrast is reported only when both endpoints admit the required timing interpretation.

For joint finite changes in initial S and I, define

$$\mathcal{I}_y(u, z) = \tau_y(u, z) - \tau_y(u, 0) - \tau_y(0, z) + \tau_y(0, 0). \tag{2.54}$$

A nonzero value indicates nonadditivity of the two timing responses. It is a finite factorial contrast and need not equal a local mixed derivative.

The perturbation in Equation (2.45) varies I with compensating variation in R. Accordingly, H3 concerns a specified direction in the initial-state space. Its support establishes a conditional dynamical mechanism, without identifying the historical cause of observed timing changes.

2.6 Analytical role of the SF-SIRS representation

The finite-dimensional form of Equation (2.8) connects transmission estimation to mechanistic analysis. For fixed calendar assignments, derivatives of β_t with respect to its coefficients are explicit. Combined with the differentiable weekly transition map, they yield trajectory derivatives through Equation (2.49) and timing derivatives through Equations (2.26) and (2.40).

SF-SIRS therefore permits the same fitted transmission function to be used for reconstruction, controlled perturbation, and mathematical characterization of peak changes. Its analytical advantage is the combination of a flexible seasonal representation with explicit finite-dimensional coordinates. This structure does not require a fixed number of transmission peaks and does not turn empirical sensitivity orderings into mathematical identities.

Chapter 3

Empirical Assessment of Peak-Shift Mechanisms in Honshu

3.1 Experimental scope and calibrated simulation baseline

All analyses in this chapter concern Honshu during 2013–2023. The experiments assess the three hypotheses formulated in Chapter 2: the mechanisms through which epidemic peaks change, the relative timing sensitivities to transmission and initial susceptibility, and the contribution of the initial infectious state.

The selected SF-SIRS fit, identified as M2 in the computational artifacts, is retained throughout the perturbation experiments. Parameters are not re-estimated after each intervention. The experiments therefore characterize responses of a fixed, calibrated dynamical system.

Two sets of evidence have distinct roles. The Phase 29 retrospective comparison assesses the choice of simulation baseline. The matched-season perturbation experiments directly assess H1–H3. Neither analysis is an out-of-sample forecast.

3.1.1 Retrospective reconstruction

We compared five internal candidates: a shared seasonal spline without historical season corrections (Spline M0), the same spline with season-specific level corrections (Spline M1), and the spline with both level and shape corrections (SF-SIRS, or M2). Two further candidates replace the shared seasonal spline with an annual Fourier function or an annual–semiannual Fourier function while retaining the M2 correction structure.

Over 2013–2023, SF-SIRS attained a mean absolute error (MAE) of 264.74 reported cases and a root mean squared error (RMSE) of 533.79 reported cases. Both errors were smaller than those of the other four candidates (Table 3.1). The comparison includes 572

weekly likelihood observations after excluding the initialized first row.

Table 3.1: Retrospective reconstruction errors for Honshu. The full-period comparison covers 2013–2023; the later-period comparison covers 2020–2023. Errors are measured in reported weekly cases.

	2013–2023		2020–2023	
Model	MAE	RMSE	MAE	RMSE
Spline M0	408.03	754.24	222.47	508.14
Spline M1	319.24	618.45	223.21	516.23
SF-SIRS (M2)	264.74	533.79	240.18	590.62
Annual Fourier with M2 correction	360.15	691.04	262.21	591.45
Annual–semiannual Fourier with M2 correction	314.67	608.02	271.15	613.76

The weekly trajectories in Supplementary Figure B.1 show where the aggregate errors arise. The left column displays the historical 2013–2019 reconstruction, while the right column enlarges 2015–2017, the period surrounding the principal timing transition. M0 smooths over several observed epidemic peaks. Adding season-specific levels improves the reconstruction; allowing season-specific shape changes further improves the alignment of the historical waves. The annual and annual–semiannual alternatives also reconstruct recurrent outbreaks, but retain larger historical count errors than SF-SIRS.

The improvement is not uniform across subperiods. For 2020–2023 alone, M0 and M1 have smaller MAE and RMSE than SF-SIRS (Table 3.1). The selection of SF-SIRS is thus based on its full-period reconstruction and its representation of the historical timing transition, rather than on a claim of superiority in every period.

Peak timing supplies a separate check on the simulator. The seasons are indexed from week 27 of one year through week 26 of the next. The observed peak advanced by 13 weeks between the 2015/16 and 2017/18 seasons. SF-SIRS reproduced an 11-week advancement, compared with 5 weeks for M0, 9 weeks for M1, 9 weeks for the annual Fourier candidate, and 8 weeks for the annual–semiannual candidate (Supplementary Figure B.3). The corresponding absolute displacement errors were 2, 8, 4, 4, and 5 weeks, respectively.

The agreement of the displacement alone does not establish accurate endpoint timing, as shown by Equation (1.3). Across the seven seasons beginning in 2013–2019, SF-SIRS and M1 both had a mean absolute peak-time error of 1.29 weeks; SF-SIRS had a smaller maximum absolute error (3 rather than 4 weeks). Supplementary Figure B.2 displays the errors by season and makes visible the comparatively large errors around 2015 in several alternative models. The 2012 season is incomplete and has no evaluated peak-time error.

Taken together, the weekly reconstruction and timing diagnostics support SF-SIRS as the selected simulation baseline for H1–H3. They do not eliminate reconstruction error or establish unique identification of its latent transmission and susceptibility trajectories.

3.1.2 Seasonal coverage and perturbation conventions

The mechanistic experiments use ten complete epidemic windows, from 2013/14 through 2022/23. Each window begins at epidemiological week 27 and ends at week 26 of the following year. The first and last partial seasons of the observation period are not counted as complete experimental seasons.

The experiments follow the saved weekly panel. In particular, the 2020/21 window contains 53 recorded observations, whereas the 2015/16 panel omits calendar week 53. Peak locations are interpreted on the recorded observation index; the seasonal transmission coordinate retains its 52-week period.

For each season, simulation restarts from the saved state immediately preceding the first affected transition. This matched-season design isolates the response to an intervention in the target season from changes accumulated under an intervention beginning in 2013.

The primary analysis uses the fitted seasonal correction without boundary blending. A boundary-blended implementation provides a numerical sensitivity analysis. These two implementations are not treated as independent datasets or statistical replicates.

3.2 Hypothesis 1: displacement and dominance exchange

3.2.1 Frequency of displacement events

Transmission was perturbed multiplicatively over the target season:

$$\beta_t(v) = \beta_t e^v, \qquad v \in [-\log(1.2), \log(1.2)]. \tag{3.1}$$

The initial state was held fixed. The resulting multiplier ranges from 1/1.2 to 1.2. Peak branches and their heights were tracked along the saved perturbation grid, with local refinement used to investigate changes in the selected maximum.

To evaluate Equation (2.31), a parent-grid timing event was counted as pure displacement only if its saved classification indicated displacement without exchange, mixed status, or unresolved status. All detected parent-grid timing events remained in the denominator. Refined subintervals were not treated as additional independent parent events.

Under the primary implementation, 72 of 80 parent-grid timing events were classified as pure displacement, giving a pooled fraction of 90.0%. Giving each of the ten seasons equal weight yielded

$$\frac{1}{10}\sum_{y=2013}^{2022}\frac{|\mathcal{D}_y|}{|\mathcal{E}_y|} = 0.7083. \tag{3.2}$$

Thus, the season-averaged majority criterion in H1 was satisfied even when mixed and unresolved events were retained as non-displacement events.

The corresponding boundary-blended analysis classified 69 of 77 events as pure displacement. Its pooled fraction was 89.61%, and its equally weighted seasonal fraction was 70.50%. The majority finding therefore persisted under this numerical sensitivity analysis.

The seasonal pattern was heterogeneous (Figure 3.1a). Every detected timing event in the four seasons from 2013/14 through 2016/17 was classified as pure displacement. The primary fractions subsequently declined to 2/3 in 2017/18, 3/4 in 2018/19, and 5/6 in both 2019/20 and 2020/21. No event was resolved as pure displacement in 2021/22 or 2022/23. In 2021/22, the sole timing event remained unresolved; its zero contribution to the conservative fraction is not evidence of a resolved alternative mechanism.

3.2.2 Dominance exchanges remain dynamically important

The predominance of displacement by event frequency did not exclude dominance exchange. Targeted branch comparisons identified signed reversals in competitor heights in 2017/18, 2018/19, and 2022/23. Under the primary implementation, the selected peak changed from seasonal position 24 to 11, from 23 to 12, and from 51 to 4, respectively (Table 3.2).

Table 3.2: Resolved dominance exchanges under the primary transmission perturbation. The bracket endpoints are values of the log multiplier v. The height gap is the early competitor's height minus the late competitor's height; its change in sign establishes a reversal of their relative heights. Peak positions are recorded seasonal indices, not calendar weeks.

Season	v_{left}	v_{right}	Peak positions	Left gap	Right gap
2017/18	−0.160600	−0.160244	$24 \to 11$	−4.49	2.50
2018/19	−0.169146	−0.168790	$23 \to 12$	−4.30	4.73
2022/23	0.066946	0.068371	$51 \to 4$	−38.57	3.02

The 47-position exchange in 2022/23 illustrates why event frequency and movement magnitude answer different questions. An infrequent exchange can produce a large change in the selected maximum. Here the change represents competition between peaks near opposite ends of the fixed seasonal window, rather than continuous movement of a single peak through 47 weeks.

H1 is therefore supported as a statement about the frequency of displacement under the specified perturbation design. The stronger statement that dominance exchange is absent, or negligible in its contribution to large peak changes, is not supported.

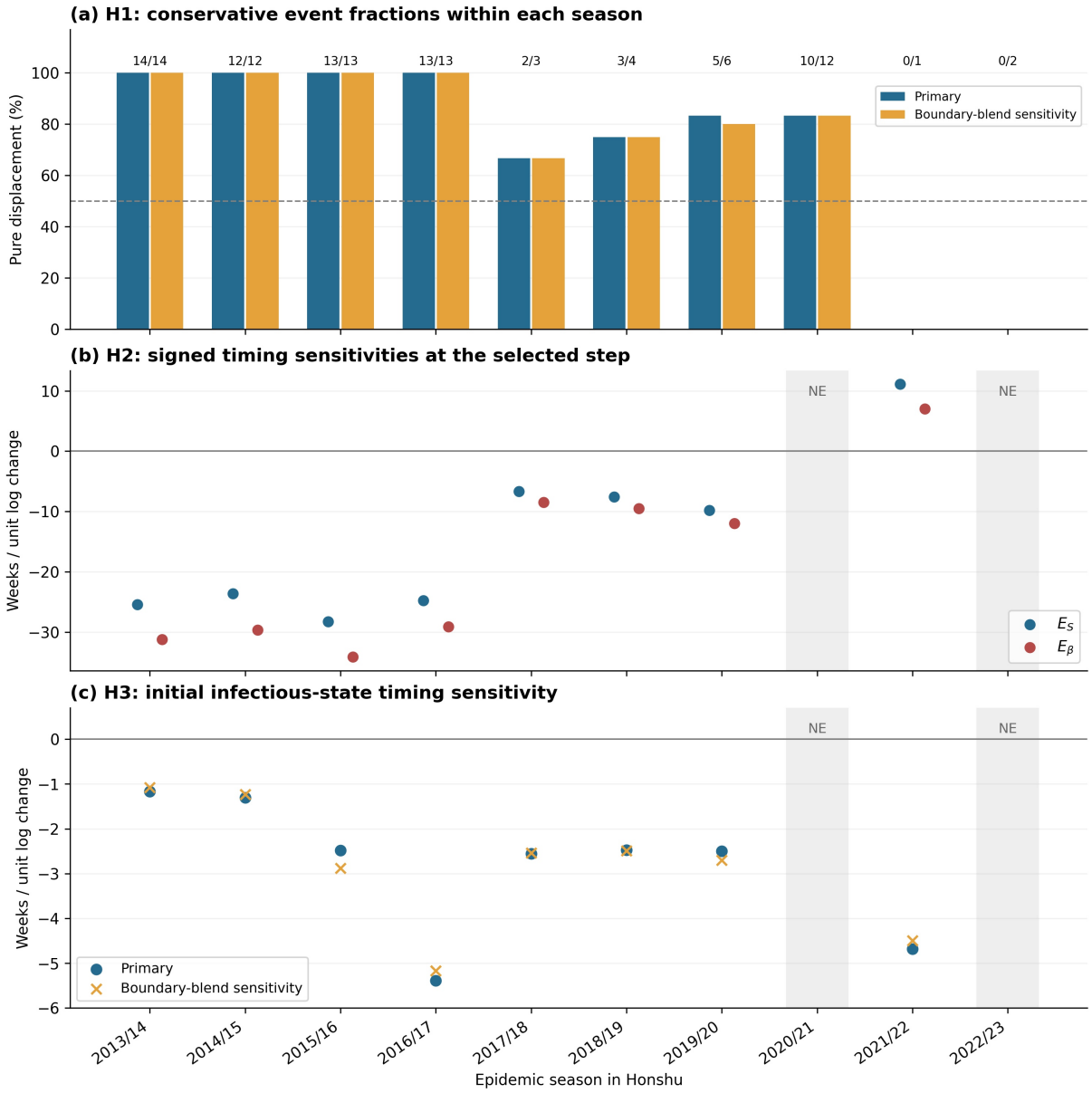


Figure 3.1: Assessment of H1–H3 across the ten complete Honshu seasons contained in 2013–2023. (a) Pure-displacement fractions retain all parent-grid timing events in the denominator, including mixed and unresolved events. Labels show primary-analysis numerators and denominators; the dashed line marks one half. (b) Signed sensitivities to initial susceptibility and transmission under the primary implementation, evaluated at the selected finite-difference step. (c) Initial infectious-state sensitivities under the primary and boundary-blended implementations. Negative sensitivities indicate advancement. Shaded seasons marked NE did not provide an eligible local estimate; missing values are not zero responses.

3.3 Hypothesis 2: relative sensitivity to transmission and susceptibility

3.3.1 Matched perturbations and numerical estimation

Initial susceptibility and transmission were perturbed according to Equation (2.36). The susceptibility intervention changed the initial susceptible state, with compensating adjustment of the recovered state. The transmission intervention multiplied the effective transmission schedule throughout the target window.

Both interventions used matched log-scale changes. Timing sensitivities were estimated by central differences of the reconstructed peak time over five step sizes,

$$h \in \left\{ \frac{\log(1.2)}{4}, \frac{\log(1.2)}{8}, \frac{\log(1.2)}{16}, \frac{\log(1.2)}{32}, \frac{\log(1.2)}{64} \right\}. \tag{3.3}$$

The saved seasonal summary selects the median eligible log-step. These numerical slopes approximate the local sensitivities defined in Equation (2.37); they are not independent statistical observations.

Eight of the ten seasons provided eligible matched estimates. The 2020/21 and 2022/23 seasons failed the saved window or peak-eligibility checks and were excluded from the local sensitivity comparison.

3.3.2 A predominant ordering with a post-pandemic exception

Transmission had the larger absolute timing sensitivity in seven of the eight eligible seasons:

$$\frac{\#\{y : |\widehat{E}_{\beta,y}| > |\widehat{E}_{S,y}|\}}{|\mathcal{Y}_{\text{elig}}|} = \frac{7}{8}. \tag{3.4}$$

The same seasonal ordering was obtained under boundary blending.

For the seven seasons from 2013/14 through 2019/20, both sensitivities were negative. Increasing either initial susceptibility or transmission advanced the reconstructed peak, with a larger absolute response to transmission (Table 3.3). At the selected steps, the ratio $|\widehat{E}_{\beta,y}|/|\widehat{E}_{S,y}|$ ranged approximately from 1.18 to 1.27 across these seasons.

The 2021/22 season exhibited the opposite ordering:

$$\widehat{E}_{S,2021} = 11.102, \qquad \widehat{E}_{\beta,2021} = 7.020. \tag{3.5}$$

Both interventions delayed the local peak, and susceptibility had the larger absolute response. This exception persisted across all four eligible step sizes and under boundary blending.

The results support H2 as a majority statement conditional on the eligible seasonal

Table 3.3: Primary seasonal timing sensitivities and finite initial-I contrasts. Sensitivities are measured in weeks per unit log change. The final column compares initial infectious states $I_{r_y}/1.2$ and $1.2I_{r_y}$, with fixed initial susceptibility and fixed transmission. NE denotes an ineligible estimate.

Season	$\widehat{E}_S$	$\widehat{E}_\beta$	$\widehat{E}_I$	$\Delta\tau_I(1.2)$
2013/14	−25.431	−31.243	−1.165	−0.417
2014/15	−23.614	−29.685	−1.301	−0.477
2015/16	−28.241	−34.128	−2.483	−0.879
2016/17	−24.772	−29.121	−5.389	−2.144
2017/18	−6.673	−8.497	−2.559	−0.964
2018/19	−7.584	−9.511	−2.472	−0.925
2019/20	−9.784	−11.986	−2.495	−0.928
2020/21	NE	NE	NE	NE
2021/22	11.102	7.020	−4.683	NE
2022/23	NE	NE	NE	NE

The selected step is $\log(1.2)/16$ except for the 2021/22 susceptibility and transmission comparison, which uses $\log(1.2)/32$. The initial-I estimate for 2021/22 uses $\log(1.2)/16$. Finite-contrast eligibility is evaluated separately from local-sensitivity eligibility.

panel. They do not support a universal inequality or a universal advancement response. Moreover, the comparison is specific to peak timing: the response of an epidemic centroid need not have the same ordering or sign.

The magnitude estimates also require attention to step-size dependence. In 2016/17, the ranges across eligible steps were 11.45 for the susceptibility slope and 21.77 for the transmission slope. Several larger perturbations changed the reconstruction stencil. The selected finite-step values should therefore not be interpreted as uniformly precise evaluations of a limiting derivative, even though the reported seasonal ordering favors transmission.

3.4 Hypothesis 3: initial infectious-state memory

3.4.1 Local responses at fixed initial susceptibility

The third experiment varied the initial infectious population according to Equation (2.45). Initial susceptibility and the transmission schedule were held fixed, and the recovered population absorbed the compensating state change. Subsequent susceptibility was allowed to evolve through the SIRS recurrence.

All eight eligible seasons had a negative estimated initial-I timing sensitivity (Figure 3.1c). Under the primary implementation, $\widehat{E}_I$ ranged from -5.389 to -1.165 weeks per unit log change. Increasing the initial infectious population therefore advanced the reconstructed local peak in every season for which this comparison was eligible.

The sign persisted under boundary blending. Across the five tested steps, the primary

initial-I slope ranges were below 0.034 weeks per unit log change in all eligible seasons. This numerical stability supports the interpretation of the detected response as a local property of the fitted trajectories.

The 2021/22 season is particularly informative. In that season, increasing initial susceptibility or transmission delayed the peak, whereas increasing the initial infectious population advanced it. The infectious-state response is therefore not a simple replication of the two responses examined under H2.

3.4.2 Finite perturbations and effect magnitude

For a finite contrast, the two initial infectious states were $I_{r_y}/1.2$ and $1.2I_{r_y}$. Their ratio is 1.44; the contrast is not a 20% increase from the lower state. The resulting peak-time difference is

$$\Delta\tau_{I,y}(1.2) = \tau_y\{\log(1.2)\} - \tau_y\{-\log(1.2)\}. \tag{3.6}$$

Seven seasons admitted an eligible finite peak-time contrast. Every primary contrast was negative, ranging from -0.417 to -2.144 weeks (Table 3.3). The largest advancement occurred in 2016/17. Under boundary blending, the corresponding advancement was 2.000 weeks.

A nonzero local response did not imply eligibility for the entire finite perturbation interval. In particular, 2021/22 provided a stable local initial-I sensitivity but failed the endpoint peak or support criteria for the finite contrast. The finite responses for 2020/21 and 2022/23 were also ineligible. These results are retained as missing comparisons, rather than interpreted as an absence of infectious-state effects.

The experiments support H3 as an additional conditional timing mechanism: changing the initial infectious state can alter peak timing while initial susceptibility and transmission are held fixed. The finite responses are generally modest over the specified perturbation range, and do not by themselves account for the full historical 13-week displacement.

3.5 Scope of the experimental evidence

The three experiments distinguish properties that would be obscured by a single summary of epidemic timing. Displacement accounts for most detected timing events under the specified grid, while dominance exchanges remain capable of producing large changes. Transmission produces the larger local timing response in most eligible seasons, but the 2021/22 exception demonstrates that this ordering depends on the dynamical state. Initial infectious-state variation provides a further timing response whose sign can differ from those of susceptibility and transmission.

These statements concern the fitted SF-SIRS system for Honshu during 2013–2023. The full-period reconstruction includes the complete observation panel, whereas the mech-

anistic comparisons use the ten complete seasonal windows and retain their numerical exclusions. No results from other Japanese regions are included.

The event fractions describe the chosen perturbation domain and grid; the seasonal sensitivity counts describe a finite set of fitted trajectories. Neither quantity is interpreted as an independent-sample probability or a population-level significance test. The experiments establish supported dynamical responses of the calibrated simulator, without assigning those perturbations as the identified historical causes of the observed transition.

Chapter 4

Comparison with the Original RSV Transmission Model

4.1 Comparison design

We compare SF-SIRS with the original Honshu SIRS model of Park et al. [5]. The comparison concerns the original Bayesian model, rather than a maximum a posteriori approximation or a later extension of that model.

Both models were evaluated on the same Honshu weekly panel covering 2013–2023. The panel contains 573 rows, of which 572 contribute to the likelihood; the initialized first row is excluded. The models use the same recorded case counts, the same evaluation periods, and the same convention for calendar week 53. Verification of the implemented SIRS transition and reported-case calculations produced zero discrepancy in the shared equation check; the discrepancy in the included negative-binomial likelihood check was 3.33×10^{-10}, below the prespecified tolerance of 10^{-7}.

SF-SIRS was estimated by penalized maximum likelihood and is represented here by its fitted expected-case trajectory. Park et al.'s model was estimated by posterior sampling. For the Bayesian model, the posterior median of the expected-case trajectories is used for the primary MAE and WAPE comparisons, and their posterior mean is used for the primary RMSE comparison. These summaries are computed pointwise from complete posterior trajectories; they are not trajectories simulated using marginal posterior mean or median parameter values.

All accuracy measures in this section describe retrospective reconstruction of observations used for estimation. They do not measure prediction of an unseen future period.

4.2 Reconstruction of reported cases

Let y_t denote observed reported cases and $\widehat{\mu}_t$ a model's summarized expected reported cases. For an evaluation set $\mathcal{T}$, we use

$$\mathrm{MAE} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} |y_t - \widehat{\mu}_t|, \tag{4.1}$$

$$\mathrm{RMSE} = \left\{ \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} (y_t - \widehat{\mu}_t)^2 \right\}^{1/2}, \tag{4.2}$$

$$\mathrm{WAPE} = \frac{\sum_{t \in \mathcal{T}} |y_t - \widehat{\mu}_t|}{\sum_{t \in \mathcal{T}} y_t}. \tag{4.3}$$

Over the full 2013–2023 panel, SF-SIRS attained an MAE of 264.74 cases, compared with 419.72 for the Park posterior median expected-case trajectory. Its RMSE was 533.79 cases, compared with 744.74 for the Park posterior mean (Table 4.1). The corresponding reductions in these two errors were 36.9% and 28.3%. SF-SIRS also had a lower full-period WAPE, 0.159 compared with 0.252 for the Park posterior median.

Table 4.1: Retrospective reconstruction on the common Honshu panel. MAE and WAPE for Park use the pointwise posterior median of expected cases; RMSE uses the pointwise posterior mean. The periods 2013–2019 and 2020–2023 partition the full evaluation panel after the initialized row is excluded.

Period	Rows	MAE		RMSE	
		SF-SIRS	Park	SF-SIRS	Park
2013–2019	363	278.88	480.63	498.14	774.50
2015–2017	156	286.04	570.84	540.37	913.02
2020–2023	209	240.18	313.93	590.62	690.01
2013–2023	572	264.74	419.72	533.79	744.74

The difference is particularly pronounced during 2015–2017, the period containing the historical timing transition. In this window, SF-SIRS attained an MAE of 286.04 cases, compared with 570.84 for Park. The advantage remains in 2020–2023, although the difference in RMSE is smaller (Table 4.1). Figure 4.1 shows the weekly observations and reconstructed expected-case trajectories. Both models reproduce the broad epidemic sequence, while their differences are most visible around individual waves and outbreak maxima.

Figure 4.2 presents the count-error comparison alongside the computational and parameter-count comparisons. The full numerical values are retained in Tables 4.1 and 4.3.

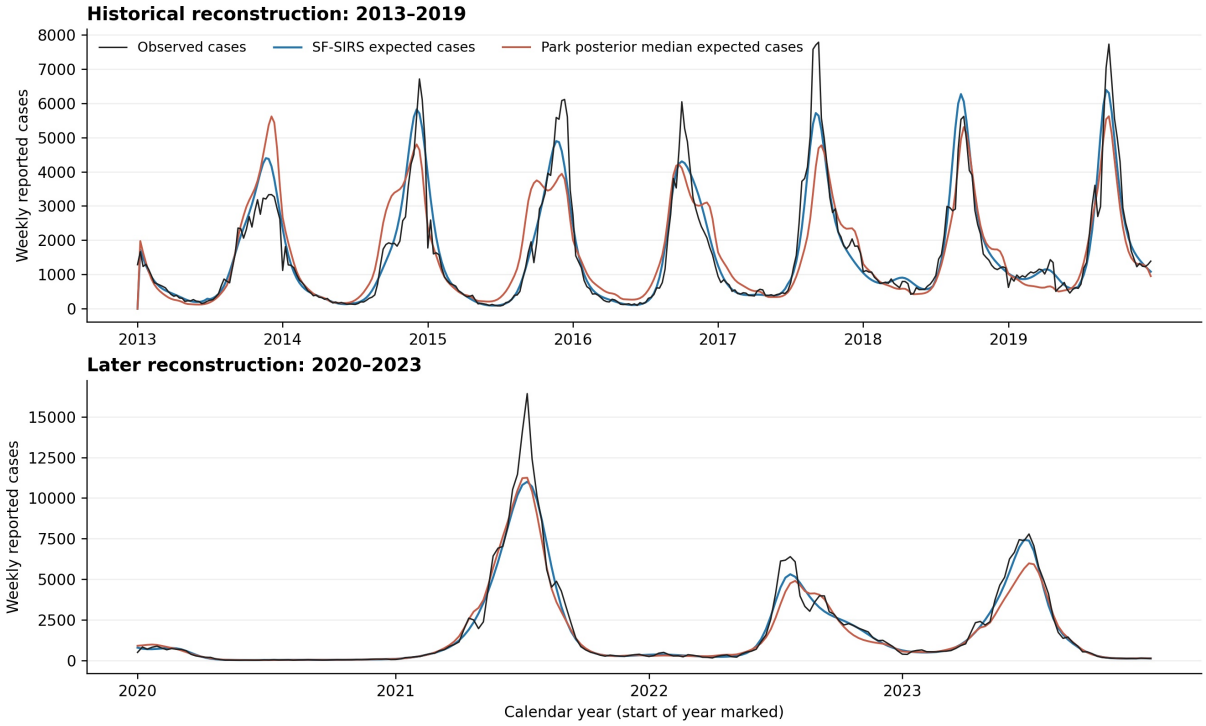


Figure 4.1: Retrospective reconstruction of reported RSV cases in Honshu. Black denotes observed weekly cases, blue the SF-SIRS expected cases, and red the pointwise posterior median of Park's expected-case trajectories. The upper panel covers 2013–2019 and the lower panel 2020–2023. The first, initialized observation is shown for temporal context but does not contribute to the likelihood. These curves are fitted reconstructions, not forecasts.

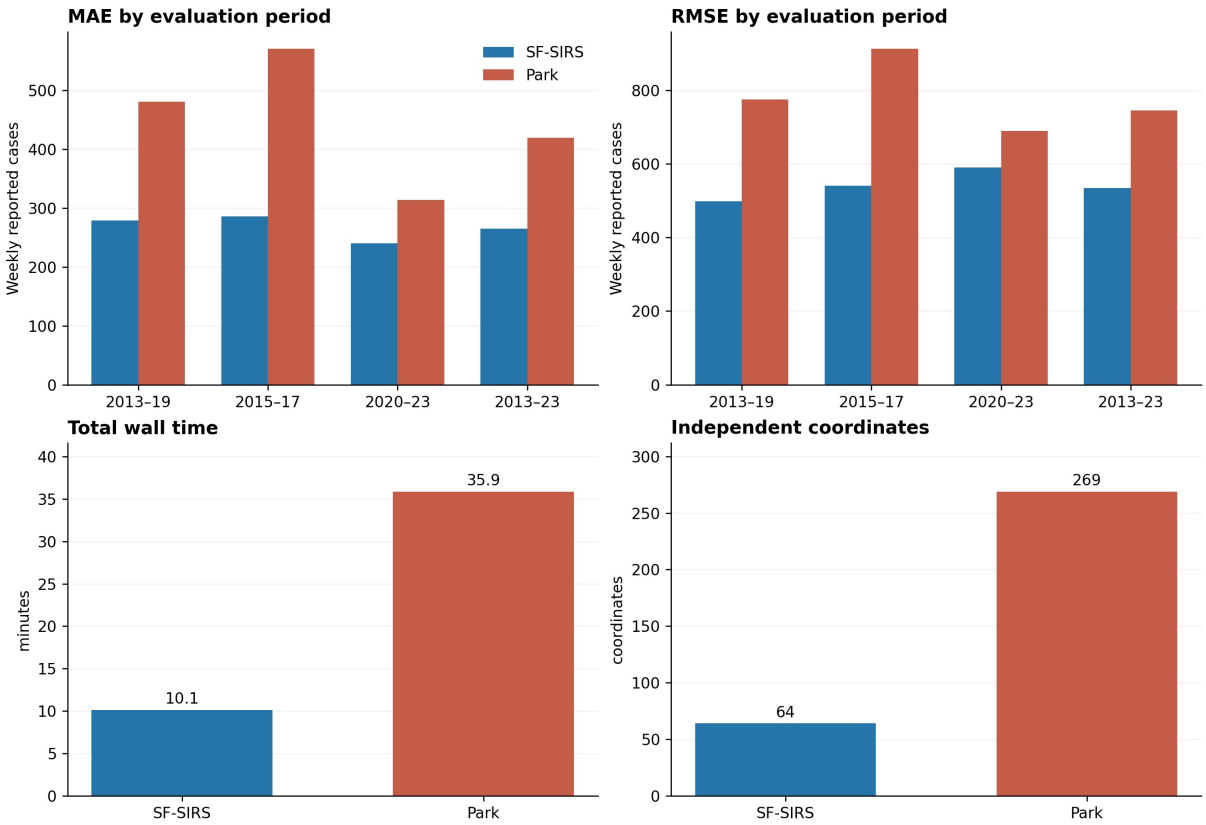


Figure 4.2: Comparison on the common Honshu panel. Upper panels show retrospective MAE and RMSE by evaluation period. Park MAE uses the posterior median expected-case summary and Park RMSE the posterior mean expected-case summary. Lower panels show the total wall time of one measured fit and the number of independent fitted coordinates. Wall time compares distinct inferential outputs and different core allocations; coordinate counts are not effective degrees of freedom.

4.3 Accuracy of epidemic timing

Case-count accuracy and peak-time accuracy are distinct properties. For the 2015/16 epidemic season, the observed peak was at seasonal position 24. SF-SIRS placed it at position 21, an error of −3 weeks, whereas the Park posterior mean and median expected-case trajectories both placed it at position 23, an error of −1 week. For 2017/18, the observed position was 11; SF-SIRS placed it at 10 and Park at 12. The two models therefore had equal absolute error of one week at this endpoint.

Both models reproduced an 11-week advancement between these seasons, compared with the observed 13-week advancement. Their displacement errors were consequently equal at two weeks. The fitted displacement alone does not distinguish their accuracy at the individual endpoints: Park reconstructed the 2015/16 peak more closely.

The epidemic center of gravity provides a complementary timing measure. For cases c_w at seasonal position w, it is defined as

$$G = \frac{\sum_w w c_w}{\sum_w c_w}. \tag{4.4}$$

The observed center of gravity moved by −4.463 weeks between the two seasons. SF-SIRS reproduced a movement of −3.268 weeks, giving an absolute movement error of 1.194 weeks. The Park posterior mean expected-case trajectory reproduced a movement of −1.388 weeks, giving an error of 3.075 weeks. Thus, SF-SIRS reproduced the shift of the epidemic's mass more closely, while Park placed the 2015/16 maximum more accurately (Table 4.2).

Table 4.2: Timing of the 2015/16 to 2017/18 transition. Peak positions are within epidemic seasons beginning in week 27. Center-of-gravity movement is based on expected-case trajectories. Park entries use the posterior mean expected-case trajectory; the posterior median gives the same two peak positions and a center-of-gravity movement of approximately −1.389 weeks.

	Observed	SF-SIRS	Park	Unit
2015/16 peak position	24	21	23	week
2017/18 peak position	11	10	12	week
Peak movement	−13	−11	−11	weeks
Center-of-gravity movement	−4.463	−3.268	−1.388	weeks

The post-pandemic seasonal peak summaries require particular care. In 2020/21 the fitted and observed maxima occur at the boundary of the recorded 53-observation window, and the 2022/23 fitted maxima occur at its boundary. An apparently small peak-position error in these seasons does not establish an interior, well-resolved epidemic peak.

4.4 Computational cost and parameter count

One timed SF-SIRS fit took 608.23 seconds, including preparation and post-processing. One Park fit took 2153.27 seconds, including 42.51 seconds of Stan compilation. Both were measured on an Intel Xeon Gold 6530 system. SF-SIRS used a single R optimizer process, whereas Park used four Stan chains allocated to four cores. The observed ratio of wall times was approximately 3.54. These are single measured fits, not estimates of mean runtime across repeated runs.

Table 4.3: Computational measurements and free-coordinate counts from Phase 28. Each runtime is one measured fit on the same CPU model. The fitted products and core allocations differ. Free-coordinate counts describe the specified parameterizations, not effective degrees of freedom.

Quantity	SF-SIRS	Park
Preparation (seconds)	0.92	1.37
Compilation (seconds)	0.00	42.51
Fitting (seconds)	607.28	2063.08
Post-processing (seconds)	0.03	46.31
Total wall time (seconds)	608.23	2153.27
Allocated cores	1	4
Shared seasonal coordinates	8	52
Time-varying correction coordinates	52	209
Initial-state coordinates	2	2
Observation coordinates	2	2
Other fitted epidemiological coordinates	0	2
Fitted prior or penalty hyperparameters	0	2
Total free coordinates	64	269

The counts reflect different allocations of model flexibility. SF-SIRS uses eight shared seasonal coordinates and 52 coordinates for its historical and later corrections. Park estimates 52 weekly seasonal transmission values and 209 later weekly correction values. The total counts are 64 and 269 independent fitted coordinates, respectively.

This reduction in coordinates makes the SF-SIRS transmission function explicit and relatively compact. It does not imply that its effective statistical complexity is exactly 64/269 of Park's: the two models impose different smoothness structures, and raw parameter counts are not effective degrees of freedom.

The runtime comparison likewise concerns different inferential products. SF-SIRS supplies a penalized likelihood point estimate for repeated deterministic perturbation experiments. Park supplies joint posterior draws and uncertainty propagation. Moreover, the Phase 28 SF-SIRS fit retains reference-derived fixed values for the importation and immunity-waning parameters, whereas Park estimates their counterparts. The recorded time advantage should therefore be interpreted as an advantage for the specific fitted simulation workflow, not as the cost of producing equivalent posterior information.

4.5 Interpretability and inferential trade-offs

Both models retain the same basic mechanistic state variables: susceptible, infectious, and recovered populations, together with an observation model for reported cases. Their principal interpretive difference is the decomposition of time-varying transmission.

SF-SIRS distinguishes a shared cyclic seasonal shape, historical season-specific level changes, historical season-specific shape changes, and a later smooth temporal correction. Its finite-dimensional form makes the relevant perturbation coordinates explicit. The analyses in Chapter 3 exploit this structure to change transmission, initial susceptibility, and the initial infectious state in controlled experiments.

Park separates a shared 52-week seasonal profile from later weekly transmission multipliers. The profile is regularized by a cyclic random-walk prior, while later multipliers are assigned their own prior structure. This model offers a direct posterior distribution over transmission trajectories and other fitted parameters. The Phase 28 sampling diagnostics passed their recorded checks: there were no divergent transitions or maximum tree-depth hits; the largest recorded $\widehat{R}$ was 1.004 and the smallest bulk effective sample size was approximately 1011.

SF-SIRS provides a point estimate and numerical optimization diagnostics rather than a posterior distribution. The posterior uncertainty available from Park is a substantive advantage when uncertainty in latent states or counterfactual trajectories is central to the inferential question.

The absolute fitted transmission levels should not be interpreted as a direct biological comparison between the two models. Transmission, the reporting multiplier, and latent initial states are jointly estimated under different regularization and nuisance-parameter policies. Moreover, neither decomposition uniquely attributes a fitted transmission change to climate, contact behaviour, reporting changes, or another unobserved cause. The interpretability advantage of SF-SIRS is the explicit definition of the interventions used in its mathematical and simulation analyses; causal attribution requires additional information.

Chapter 5

External Forecast Evaluation in Honshu, 2024–2026

5.1 Model and forecast design

We evaluated the M2 specification of the Spline–Fourier SIRS (SF-SIRS) model described in Chapter 2. M2 combines a cyclic spline for baseline seasonal transmission with season-specific level and Fourier shape adjustments in the historical period and a smooth transmission correction after 2019. Weekly RSV reports are linked to modelled incidence by a reporting fraction and a negative-binomial observation model. The fit was obtained by penalized maximum likelihood using the 2013–2023 Honshu series.

The forecast was initialized at the fitted susceptible, infectious, and recovered counts in 2023 week 52. The fitted seasonal transmission profile was repeated, and the nonseasonal transmission multiplier was held at its value at the forecast origin. The state trajectory was propagated continuously through 2026 week 53, which used the week 52 seasonal transmission value. The sum of the three compartment counts at the forecast origin was used as the fixed population denominator in the primary forecast. Point forecasts were obtained from the fitted observation mean. All post-2023 observations were reserved for evaluation. The comparator was a seasonal naïve forecast that repeated the 2023 Honshu report count for the same epidemiological week. This design evaluates a retrospective fixed-origin forecast with a specified rule for future transmission.

5.2 External surveillance data

The outcome was the weekly sum of reported RSV cases from paediatric sentinel sites across the 34 prefectures assigned to Honshu in the study data. Observations for 2024 were taken from Table 8-1 of the finalized National Epidemiological Surveillance of Infectious

Diseases (NESID) annual report.[1] Observations for 2025 and for weeks 1–44 of 2026 were drawn from the provisional Infectious Diseases Weekly Report (IDWR) tables released for 2025 week 52 and 2026 week 44, respectively.[2] All 148 evaluated weeks contained reports for the 34 prefectures. The 2025 and 2026 counts are subject to surveillance revisions.

The national selection criteria for sentinel sites changed in 2025 week 15, and the paediatric network decreased from approximately 3,000 to 2,000 sites nationwide.[3] We therefore report the two 2025 intervals separately. Comparisons from week 15 onward are diagnostics of raw report counts under the revised surveillance system.

5.3 Forecast accuracy and peak timing

Table 5.1 gives weekly mean absolute error (MAE), the week of maximum reports, and cumulative bias over each scored interval. Relative MAE improvement is $100(1 - \mathrm{MAE}_{\mathrm{M2}}/\mathrm{MAE}_{\mathrm{naive}})$. Cumulative bias is the difference between the predicted and observed totals divided by the observed total, expressed as a percentage.

Table 5.1: Fixed-origin forecast accuracy for weekly RSV reports in Honshu. Peak weeks refer to the maximum within each evaluation interval. Positive MAE improvement favours M2.

	MAE (reports/week)		Improvement	Peak week			M2 bias
Interval	M2	Naïve	(%)	Obs.	M2	Naïve	(%)
2024 W01–W52	1,157.5	1,132.0	−2.3	16	32	27	−9.3
2025 W01–W14	712.7	1,470.6	51.5	11	14	14	−22.0
2025 W15–W52	526.6	2,629.0	80.0	37	38	27	+36.6
2026 W01–W44	643.3	2,187.3	70.6	41	37	27	+55.0

In 2024, M2 had a 2.3% higher weekly MAE than the seasonal naïve forecast. The observed series reached 4,657 reports in week 16, when M2 predicted 835 reports. The M2 maximum occurred 16 weeks after the observed maximum. Its smaller absolute annual total error coincided with substantial underprediction in the first half of the year and overprediction in the second (Figure 5.1).

In the first 14 weeks of 2025, before the sentinel redesign, M2 reduced weekly MAE by 51.5% and had a smaller absolute error in 12 of 14 weeks. M2 predicted 22,621 reports over this interval, compared with 28,999 observed and 8,515 predicted by the baseline.

[1] Japan Institute for Health Security (JIHS), 2024 NESID annual tables, Table 8-1. https://id-info.jihs.go.jp/surveillance/idwr/annual/2024/syulist/index.html.

[2] JIHS, IDWR provisional weekly sentinel data. 2025 week 52 CSV; 2026 week 44 CSV.

[3] JIHS, acute respiratory infection surveillance report for 2026 week 5, notes to Figure 1B. https://id-info.jihs.go.jp/surveillance/idss/target-diseases/acute-respiratory-infection/sentinel-sites/2025/ARI_2026w05.pdf.

Both forecast maxima occurred at the evaluation boundary in week 14, three weeks after the observed maximum. The improvement in this interval was in weekly count accuracy.

From week 15 onward, the 2025 observed and M2 maxima were 3,031 reports in week 37 and 3,021 in week 38, respectively. M2 nevertheless overpredicted total reports over that interval by 36.6%. Across the full calendar year, the largest observed weekly count occurred in week 11, 27 weeks before the M2 maximum. The one-week error applies to the later interval (Figure 5.2).

Through week 44 of 2026, M2 captured the autumn rise and placed its maximum four weeks before the observed maximum, compared with 14 weeks for the baseline. The M2 maximum was 4,806 reports, 74.6% higher than the observed interval maximum of 2,752. Its cumulative prediction was 72,053 reports compared with 46,474 observed (Figure 5.3).

5.4 Limitations and directions for further work

The results establish performance against a single seasonal baseline at one fixed forecast origin. The later count comparisons combine forecasting error with changes in surveillance coverage and site composition. Honshu-specific sentinel denominators were unavailable for adjustment. The evaluation uses point forecasts, so predictive interval coverage and calibration remain unassessed. The alternative population denominator changed any weekly forecast by at most 1.42 reports. The study did not evaluate Park et al.'s model on this external panel.

A first extension is to obtain weekly counts of reporting paediatric sites by prefecture and incorporate them as exposure in the observation model. Overlapping site-level data, if available, would help assess the effect of changes in site composition. Rolling-origin tests could then evaluate shorter horizons and compare the frozen transmission correction with alternatives that allow it to evolve. Comparing the models under matched forecast origins, against both seasonal and recent-count baselines, would show whether the accuracy gain persists across forecast settings. Propagating parameter, latent-state, and observation uncertainty would allow assessment of predictive calibration for both weekly counts and peak timing.

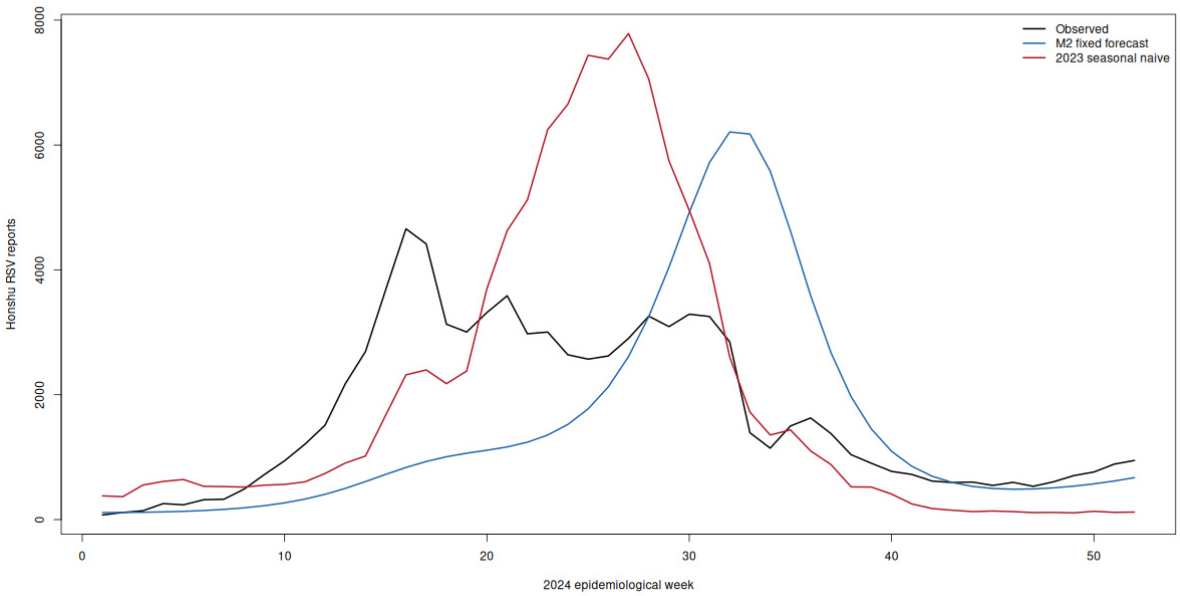


Figure 5.1: Weekly RSV reports in Honshu in 2024. The black line is the observed count, blue is the M2 point forecast, and red is the seasonal naïve forecast based on 2023.

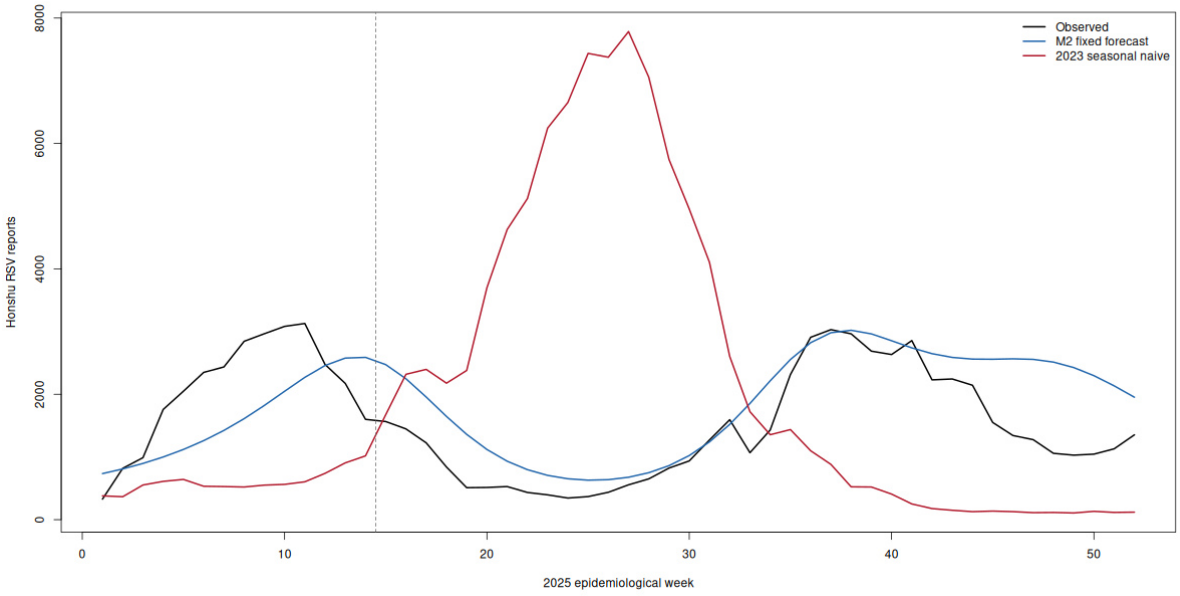


Figure 5.2: Weekly RSV reports and forecasts in Honshu in 2025. The vertical marker denotes the change in sentinel surveillance at week 15. Colours follow Figure 5.1.

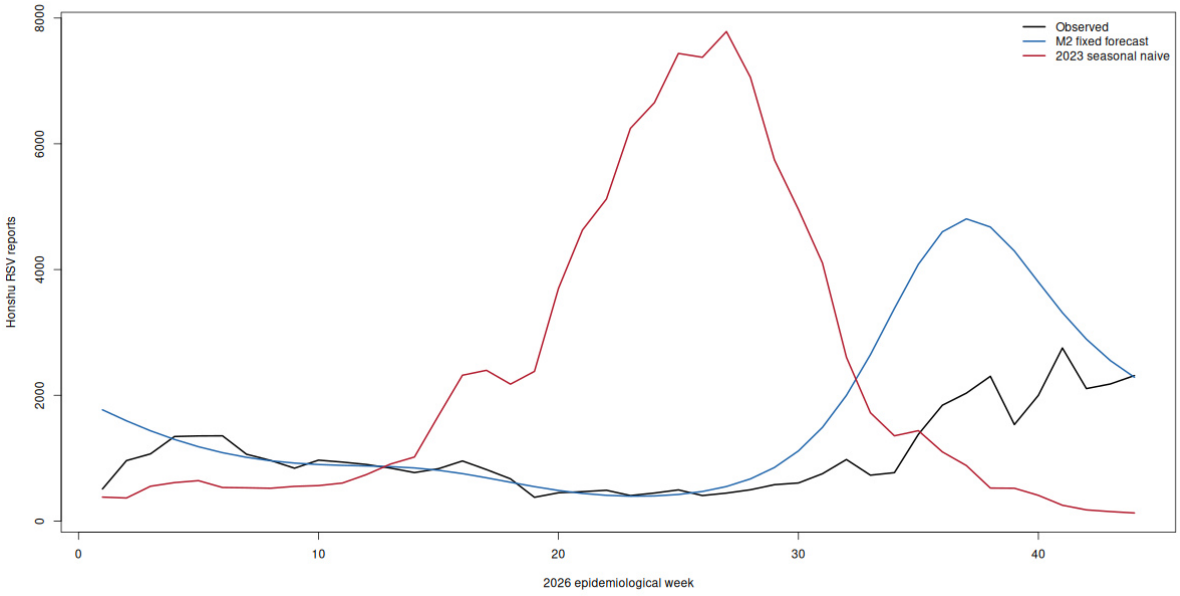


Figure 5.3: Weekly RSV reports and forecasts in Honshu in 2026. The plot and evaluation end at week 44. Colours follow Figure 5.1.

Appendix A

Proofs for the SF-SIRS Peak-Timing Framework

A.1 Differentiability of the weekly trajectory

For fixed basis functions and calendar assignments, Equation (2.8) is differentiable in its free coefficients. Its normalizing denominator is strictly positive.

Because $\mu > 0$, the denominators $\lambda_t + \mu$, $\gamma + \mu$, and $\delta + \mu$ are positive on the feasible parameter domain. The weekly transition map is therefore differentiable in the state and parameters. On a finite time horizon, repeated composition establishes differentiability of X_t and J_t with respect to every feasible perturbation considered in the main text. This statement concerns parameter derivatives and does not require calendar-time differentiability at season boundaries.

A.2 Proof of Proposition 1

Set $G(t, p) = f_t(t, p)$. At (τ_0, p_0),

$$G(\tau_0, p_0) = 0, \qquad G_t(\tau_0, p_0) = f_{tt}(\tau_0, p_0) < 0.$$

The implicit function theorem gives a locally unique differentiable function $\tau(p)$ satisfying $G(\tau(p), p) = 0$. Differentiation yields

$$f_{tt}(\tau(p), p)\tau'(p) + f_{tp}(\tau(p), p) = 0,$$

which proves Equation (2.26). Continuity preserves negative curvature locally.

For global selection, additionally assume a fixed compact window, joint continuity, and a positive height gap between this peak and all points outside a sufficiently small neighborhood. The gap persists for sufficiently small perturbations. The locally continued

peak then remains the global maximizer. □

A.3 Proof of Proposition 2

Define $D(p) = H_1(p) - H_2(p)$. By assumption,

$$D(p_0) = 0, \qquad D'(p_0) \neq 0.$$

Hence

$$D(p) = D'(p_0)(p - p_0) + o(|p - p_0|),$$

so D has opposite signs on the two sides of p_0. The positive gap over all other candidates persists locally. Thus one branch is globally dominant immediately before the crossing and the other immediately after it. Since their locations are distinct at p_0, the selected peak has distinct one-sided limits.

Moreover,

$$H_j'(p) = f_t(\tau_j(p), p)\tau_j'(p) + f_p(\tau_j(p), p) = f_p(\tau_j(p), p),$$

because $f_t = 0$ on a peak branch. Therefore the crossing condition can be evaluated through the difference in the branches' height responses. □

A.4 Proof of Proposition 3

Apply Proposition 1 separately to u and v at the same baseline peak. Both derivatives have denominator $f_{tt} < 0$, which proves Equation (2.39).

For a fixed quadratic stencil,

$$\tau = k + \frac{A}{2B}.$$

Differentiation with respect to q gives

$$\partial_q \tau = \frac{BA_q - AB_q}{2B^2}.$$

This proves Equation (2.40). If the selected stencil or branch changes, the calculation applies only on each eligible local segment. □

A.5 Proof of Proposition 4

Differentiating the initial-state perturbation gives

$$\frac{d}{dz}\begin{pmatrix} S_{r_y} \\ I_{r_y}e^z \\ M_{r_y} - S_{r_y} - I_{r_y}e^z \end{pmatrix}\Bigg|_{z=0} = \begin{pmatrix} 0 \\ I_{r_y} \\ -I_{r_y} \end{pmatrix}.$$

The chain rule applied to $X_t(z) = F_t(X_{t-1}(z))$ yields $Z_t = D_X F_t Z_{t-1}$. When q also enters F_t explicitly, the chain rule adds $\partial_q F_t$, establishing the general tangent recursion.

The peak-time formulas follow from Proposition 1 or direct differentiation of the quadratic reconstruction. They establish the route through which initial-state variation can affect timing, but do not imply that the derivative is nonzero or has a universal sign. □

A.6 Immediate incidence response and its distinction from timing

At the first affected transition, define

$$a(\lambda) = \frac{\lambda}{\lambda+\mu}\left(1 - e^{-(\lambda+\mu)}\right), \qquad J = S_{r_y} a(\lambda).$$

For $\lambda \geq 0$,

$$a'(\lambda) = \frac{\mu}{(\lambda+\mu)^2}\left(1 - e^{-(\lambda+\mu)}\right) + \frac{\lambda}{\lambda+\mu}e^{-(\lambda+\mu)} > 0.$$

Under the initial-I perturbation,

$$\frac{\partial \lambda}{\partial z}\bigg|_{z=0} = \frac{\beta I_{r_y}}{N}.$$

Thus, when $S_{r_y} > 0$, $I_{r_y} > 0$, and $\beta > 0$,

$$\frac{\partial J}{\partial z}\bigg|_{z=0} > 0.$$

An increased initial infectious population therefore increases incidence at the first affected transition. Peak timing depends instead on the subsequent trajectory and, locally, on f_{tz}. A positive immediate incidence response alone does not determine the sign of $E_{I,y}$.

The propositions prove the structural identities used to test H1–H3. Their empirical frequency, magnitude, and seasonal generality require the corresponding Honshu experiments.

Appendix B

Supplementary

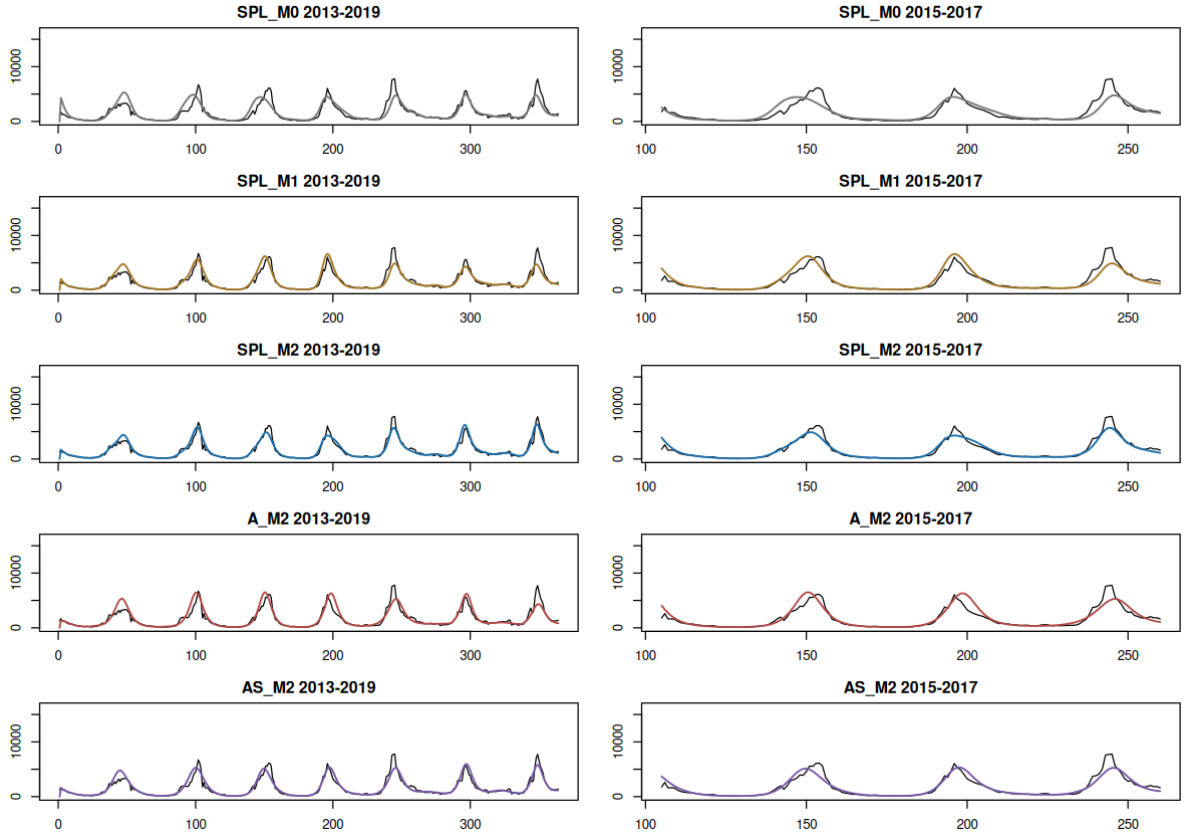


Figure B.1: Retrospective weekly case reconstruction for the five internal candidate models in Honshu. Rows correspond to Spline M0, Spline M1, SF-SIRS (Spline M2), annual Fourier with M2 corrections (A_M2), and annual–semiannual Fourier with M2 corrections (AS_M2). Black curves show reported cases; coloured curves show fitted expected reported cases. The left column covers 2013–2019, and the right column enlarges 2015–2017. The horizontal axes use the observation index of the common weekly panel. These are retrospective reconstructions, not forecasts.

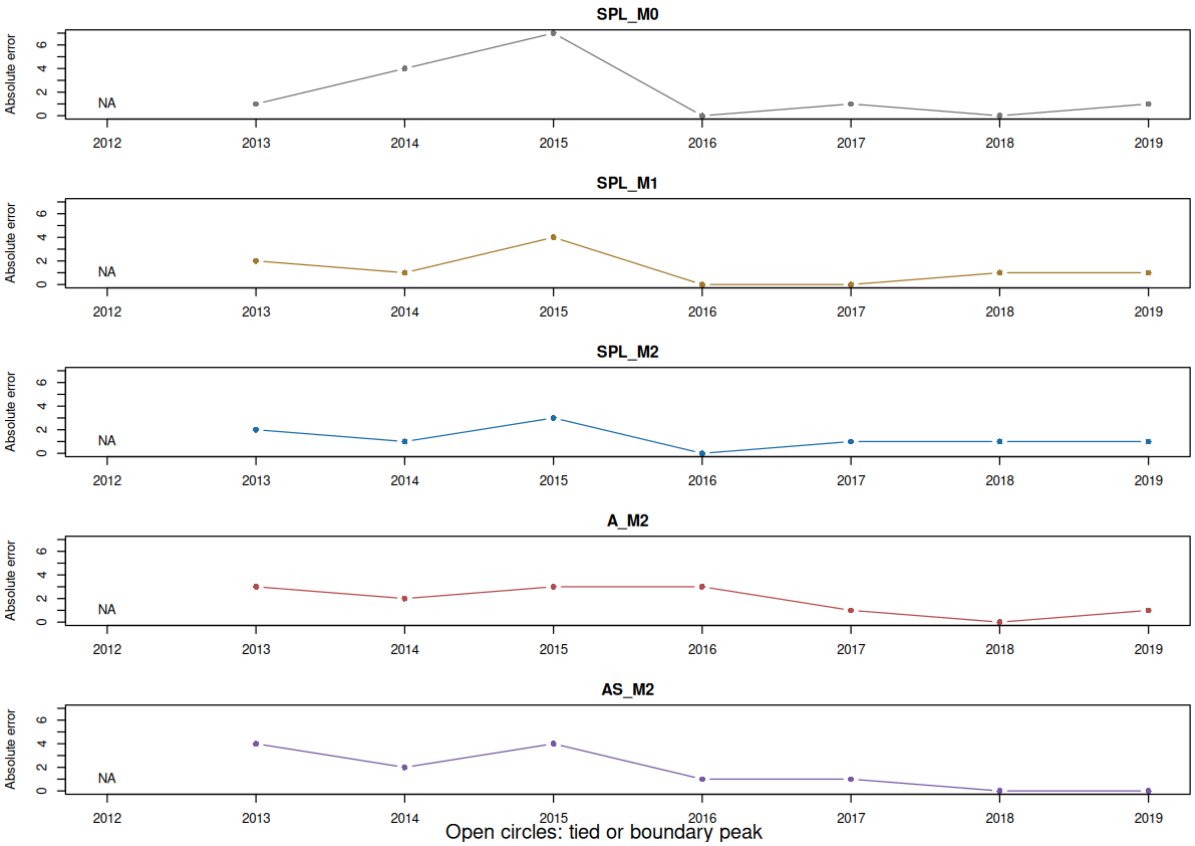


Figure B.2: Absolute error in the reconstructed epidemic peak position by season and candidate model. Seasonal years denote windows beginning in week 27 and ending in week 26 of the following year. The incomplete 2012 season is marked "NA" and excluded from the timing summary. The annotation concerning open circles identifies the plot's convention for tied or boundary peaks; such cases require separate interpretation from an unambiguous interior maximum.

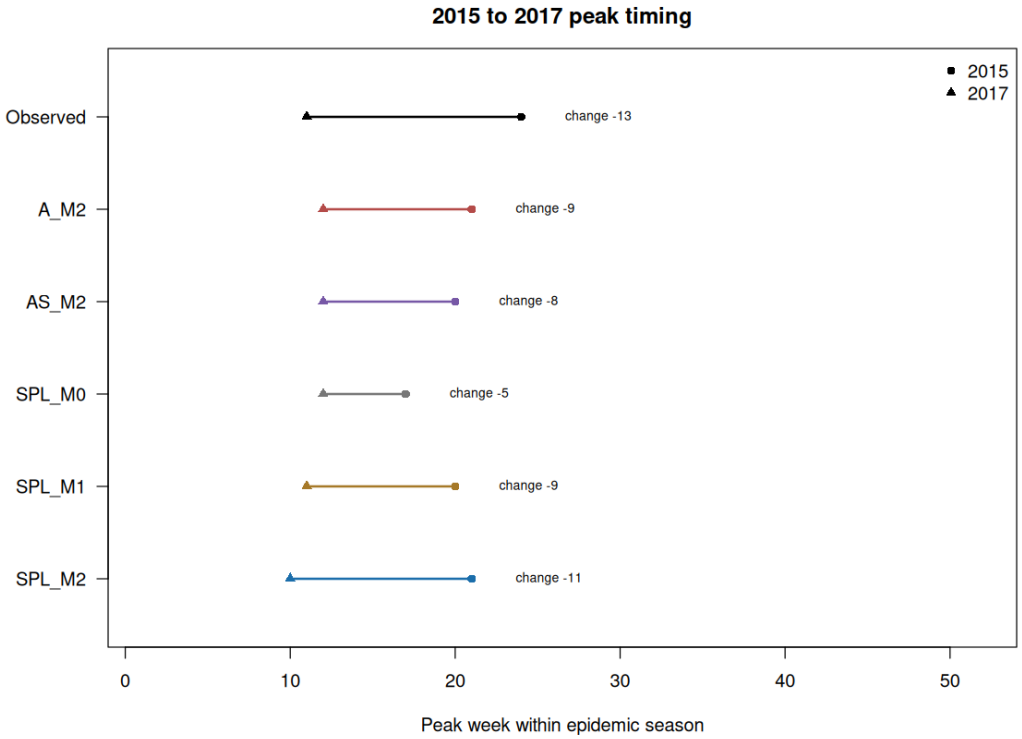


Figure B.3: Retrospective peak positions for the 2015/16 and 2017/18 epidemic seasons. Circles indicate the 2015/16 peak and triangles the 2017/18 peak; horizontal segments connect each model's two positions. The observed peak advanced by 13 weeks. SF-SIRS reproduced an 11-week advancement, while Spline M0, Spline M1, A_M2, and AS_M2 reproduced advancements of 5, 9, 9, and 8 weeks, respectively. The plotted positions are weeks within an epidemic season, not calendar-year weeks.

Bibliography

[1] Earn, D. J. D., Rohani, P., Bolker, B. M., and Grenfell, B. T. (2000). A simple model for complex dynamical transitions in epidemics. *Science*, 287(5453), 667–670. https://doi.org/10.1126/science.287.5453.667

[2] White, L. J., et al. (2007). Understanding the transmission dynamics of respiratory syncytial virus using multiple time series and nested models. *Mathematical Biosciences*, 209(1), 222–239. https://doi.org/10.1016/j.mbs.2006.08.018

[3] Pitzer, V. E., et al. (2015). Environmental drivers of the spatiotemporal dynamics of respiratory syncytial virus in the United States. *PLoS Pathogens*, 11(1), e1004591. https://doi.org/10.1371/journal.ppat.1004591

[4] Baker, R. E., et al. (2019). Epidemic dynamics of respiratory syncytial virus in current and future climates. *Nature Communications*, 10, 5512. https://doi.org/10.1038/s41467-019-13562-y

[5] Park, S. W., et al. (2025). Interplay between climate and childhood mixing can explain a sudden shift in RSV seasonality in Japan. *Nature Communications*, 16, 11385. https://doi.org/10.1038/s41467-025-66184-y

[6] Eilers, P. H. C., and Marx, B. D. (1996). Flexible smoothing with B-splines and penalties. *Statistical Science*, 11(2), 89–102. https://doi.org/10.1214/ss/1038425655

[7] Ramsay, J. O., Hooker, G., Campbell, D., and Cao, J. (2007). Parameter estimation for differential equations: A generalized smoothing approach. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(5), 741–796. https://doi.org/10.1111/j.1467-9868.2007.00610.x

[8] Bouman, J. A., et al. (2024). Bayesian workflow for time-varying transmission in stratified compartmental infectious disease transmission models. *PLoS Computational Biology*, 20(4), e1011575. https://doi.org/10.1371/journal.pcbi.1011575